GEORGY ELIAS ROY

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Data Engineer and critical thinker well adapted to working under pressure.mentored and built projects for 15+ successful clients.Proficient in data engineering, data analysis and Full Stack(MERN). Seeking opportunities in companies that value creativity and innovation.

SKILLS

Languages & tools	Python, R, SQL, JavaScript
Tools	Pandas, Numpy, Matplotlib, Seaborn, Git & GitHub, Docker, Pyspark,Plotly.js, LangChain, Tableau, PowerBI
Skills	Data Cleaning, Data Transformation, Machine Learning, Deep Learning, Data Imputation, Data Visualization, Data Modeling, Text Analysis, Hyperparameter Tuning,AI, AI Optimization Techniques,Retrieval Augmented Generation.

PROJECTS

Stock Market Visualizer webapp *Personal Project* — Apr 2024 - Present

- Built an open-source dashboard enabling interactive stock market analysis and AI-powered financial Q&A.
- Utilized plotly.js in react to create visualizations enabling interactivity by default with less coding.
- Employed FastAPI for backend. Retrieved multiple Stock Data from different sources, Performed ETL methodologies to create new data from existing ones and create a new Data Pipeline accessible via another API.
- Containerized phi-2 AI model via Ollama which has enough literacy to answer basic questions in finance keeping the scope of adding more powerful models in the future. RAG is being added to the model to improve responses(still under work)
- Tools and Libraries Used: Plotly.js, Alpha Vantage, Python & FastAPI, React.js, GitHub, Phi2, ollama,tensorflow,LangChain

Analysis of UK GDP – *Dissertation* — Jan 2023 - Aug 2023

- Deployed a project to analyze the UK GDP pre and post Brexit and forecast trends for 2023, addressing fluctuations and Important factors.
- Utilized MS Excel, Pandas, and NumPy for Extract Transform and Load(ETL).
- Employed standard data transformation and data modeling methods like predictive imputation to handle missing values and quadrant analysis to eliminate outliers to create a Data Pipeline that provide clean data.
- Observed -0.72 correlation between Nominal GDP and Gross Domestic savings rate Indicating high inflation. This holds true for the UK in recent times.
- Identified several fluctuations in Real GDP indicating unstable nature. Found factors like imports, exports etc. that directly affect GDP values and also to the extent which they affect GDP. Concluded that post brexit, UK is going through inflation and slow GDP growth.
- Tools Used: Python(NumPy, Pandas,Scikit-Learn„ Matplotlib, Seaborn, Power BI, MS Excel

Breast Cancer Analysis Using Computer Vision and Deep Learning *Personal Project* — Dec 2023 - February 2024

- Developed a project to detect breast cancer on the BreakHIS dataset using Both Machine Learning and Deep Learning.
- Implemented standard image processing techniques, including resizing, normalization, zooming, rotation, and flipping, ensuring prediction quality and consistency in the Data Pipeline. These techniques further elevated the accuracy by 7-8%.
- Achieved high accuracy (>80%) in deep learning models, with CNN and VGG16 models attaining an accuracy score of 87%(state of the art= 99.99%).
- Tools and libraries used: Python (Keras, TensorFlow), deep learning & machine learning (CNN, ANN), data manipulation (Numpy), Image Processing (OpenCV)

EXPERIENCE

Freelance, *Developer and Consultant* — 2023 - Present

- delivered projects across diverse fields including finance, Bio Informatics, Image Processing and weather data.
- Mentored over 10 data science students through personalized consultancy, guiding them in mastering Data Science. Students have 100% success rate after consultancy.
- Ensured efficient project management by utilizing GitHub for version control and maintaining comprehensive documentation.

EDUCATION

University of Essex -*MSc Data Science(Merit)* — 2022-2023
APJ Abdul Kalam Technological University -*B.Tech Computer Science and Engineering* — 2016-2020